

Smart Analytics Tool

Project Installation & Execution Guide

Step 1: Open Google Colab

Open Google Colab and create a new notebook. This environment will be used to build and run the Streamlit application.

Step 2: Install Required Libraries

Run: `!pip install streamlit pandas numpy plotly pyngrok`

Step 3: Create the Streamlit Application

Create a file named `app.py` using `%%writefile app.py` and paste the project code.

Step 4: Run the Streamlit Application

Execute: `!streamlit run app.py &>/content/log.txt &`

Step 5: Generate Public URL

Use Pyngrok to generate a public URL: `from pyngrok import ngrok public_url = ngrok.connect(8501)`

Step 6: Open the Application

Click the generated URL to open the dashboard in a browser.

Step 7: Upload Dataset

Upload the Life Expectancy Data CSV file using the file uploader.

Step 8: Explore the Dashboard

View dataset preview, missing values, statistical summary, and interactive visualizations.

Project Features

• CSV Upload • Dataset Preview • Missing Value Analysis • Statistical Summary • Histogram • Scatter Plot • Line Chart • Interactive Data Exploration

Expected Output

The application displays uploaded data, detects missing values, generates statistical summaries, and provides dynamic visualizations for better understanding of the dataset.

Submission Files

1. `app.py` 2. Life Expectancy Data.csv 3. `requirements.txt` 4. `README.pdf` 5. `Installation Guide.pdf`